

```
#include <reg51.h>

void main(void)
{
    unsigned int i;

    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444,
0xABCD};

    unsigned long sum = 0;
    for(i = 0; i < 5; i++)
    {
        sum = sum + array[i];
    }

    P0 = (unsigned char)(sum & 0xFF);
    P1 = (unsigned char)((sum >> 8) & 0xFF);
    P2 = (unsigned char)((sum >> 16) & 0xFF);
    while(1);
}
```

D:\embedded lab\exp-13.uvproj - uVision

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Registers Disassembly

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
sp	0x15
sp_max	0x15
dptr	0x0000
PC	0x009...
status	389
sec	0.0004...
psw	0x00

```

3: void main(void)
4: {
5:     unsigned int i;
6:     unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
7:     unsigned long sum = 0;
8:
9:     for(i = 0; i < 5; i++)
10:    {
11:        sum = sum + array[i];
12:    }
13:
14:    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
15:    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
16:    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
17:
18:    while(1); // Keep output stable
19:}

```

Parallel Port 0: P0: 0xCC Pins: 0xCC

Parallel Port 1: P1: 0xAB Pins: 0xAB

Parallel Port 2: P2: 0x01 Pins: 0x01

Parallel Port 3: P3: 0xFF Pins: 0xFF

Command: Running with Code Size Limit: 2K
Load "D:\embedded lab\Objects\exp-13"

Call Stack + Locals

Name	Location/Value	Type
MAIN	C:0x009	

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE

Simulation tt: 3325.52952474 sec Li6 C1 CAP. NUM SCRL OVR. R/W